Primates

22

Researchers are divided as to exactly when the primates first appeared on Earth. Some believe they evolved from an insectivorous ancestry during the late Cretaceous (135–70 million years ago) period. Others date the first primates at the Palaeocene (65–54 million years ago) period. Either way, the earliest primates were here a long time ago.

1 Define *insectivore*.

Primate groups

There are two divisions of primates. They are the *prosimians* (*Prosimii – pro* before, *simian* apelike) and the *anthropoids* (*Anthropoidea*, resembling man). There are five families of the older prosimii, or 'Old World' primates, most of which are found in Africa, the Philippines and Madagascar. Examples of prosimians are the lemur, tarsier (related to the lemur), loris, and the tree shrew.

Primates *are animals, which belong to the mammalian order primates.*

The anthropoids include the monkeys, apes and humans. Their faces are generally much flatter than the prosimians and they have two premolar teeth. Anthropoids also have areas on their faces and/or bodies that are not covered by hair. They can sit in an upright position – or at least on their haunches (the area consisting of the thighs and buttocks). Anthropoids use their opposable thumbs and grasping hands to hold objects and aid in climbing. Depending on the species, they can walk either completely, or for short periods, using bipedal (two-legged) motion. Apes lack a tail.

The 'New World' anthropoids include the monkeys and marmosets. The monkeys and marmosets are mostly arboreal and lack opposable thumbs, tending to use their prehensile tail to grasp (*prehensile* to seize). New World anthropoids walk using their arms and legs (quadrapedal). They have three premolars, as opposed to the pair found in anthropoids.

Important note: *We often describe primates as 'Old World' or 'New World'. The terms 'old' and 'new' refer to the landmasses on which the species are found. It does not refer to the evolution of the species. The Old World is the landmasses we have known about longer, including Africa and Asia. The New World, discovered much later, is the Americas.*

Primate evolution

Primates have evolved with time. As the arboreal (tree-dwelling) prosimians evolved, their eyes moved from the side to the front of the head, giving them binocular vision. This means that they can judge distance and depth more accurately. They also developed hands with separate fingers and opposing thumbs that could grasp branches. Some prosimians, such as the ring-tailed lemur, found in Madagascar, travel on all fours, with knuckles of the hands touching the ground. Other lemurs are arboreal. The big-eyed loris and the tarsier of Asia have large eyes that are set in a fixed position. These animals must turn their heads (by up to 180°) to look to the side.

2 Name two examples of primates that have a terrestrial (for the most part at least) rather than arboreal lifestyle.

3 State an advantage of binocular vision.

4 State the meaning of *arboreal*.

5 Explain why opposable thumbs are so useful.

6 What is meant by *bipedal motion*?

Research

7 Scientists do not always agree as to whether the insectivorous tree shrew should be classified as a primate. Research the features of the tree shrew. Do you agree or disagree that it should be classified as a primate? Give reasons why.

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Monkeys

Monkeys first appeared in Africa about 30 million years ago. They were a more advanced type of primate, and were more successful in competing for living space and food than lemurs, and began to take over from the lemur. (Since Madagascar had become an island by then, monkeys did not reach there. Hence the lemurs are more numerous on Madagascar.)

During their evolution primates have become larger. New World monkeys tend to be smaller than Old World monkeys. Many have a prehensile tail that acts as a useful fifth limb. In the Old World some monkeys, such as the baboon, have left the trees and become almost completely terrestrial, ie stay on the ground. The slower moving orang-utans (*'the old man of the forest'*) tend to stay in trees. The exceptions are older and larger males who travel on the ground.

8 Explain how a prehensile tail can be useful for a monkey.

The gibbons of Southeast Asia swing and leap rapidly through trees using their long arms. When on the ground they can walk upright, using their arms for balance.

Africa is home to the chimpanzee and the gorilla. The gorilla is the largest primate with an average weight of 270 kg. Both chimpanzees and gorillas climb trees, especially to nest at night, but are more terrestrial than Asian apes. All of the *great apes* (orang-utans, chimpanzees and gorillas) are able to walk upright but more often travel on all fours.

9 Explain why some primates who generally live on the ground during the day, might nest in trees at night.

All monkeys have eyes at the front of the head, giving them binocular vision. Having excellent colour vision, they can see well through vegetation. The characteristics gained from living in trees have been a great help to those monkeys now living on the ground. These skills include good eyesight, an ability to learn quickly from experience and manipulative hands.

10 What is meant by *manipulative hands*?

As primates evolved their bodies grew in size and so did their brains. Approximately 3 million years ago the first *hominids* (apes resembling humans) appeared. The shapes of the hips and pelvis suggest that they walked upright on two legs.

Primate social patterns

Primates exhibit a wide range of social patterns. They can communicate through sounds, facial expressions, making signals and posturing. Some, like the gibbons, live in family groups. Howler monkeys and chimpanzees form large, loosely organised groups. A typical orang-utan group has one or two females with young, with males joining them only for mating. Gorillas live in family units. Different species look after their offspring for different lengths of time. Tree shrews need to care for their young for approximately 3 months, great apes for about 7 years and humans need even longer periods of time.

11 Explain what is meant by *posturing*.

12 Discuss the statement: *'The more evolved the organism, the longer the developmental time of its offspring'.*

The *Apes,* including the gibbon, chimpanzee, orang-utan and gorilla (*Pongidae*), have broad chests, arms longer than their legs and lack tails. They are larger and more intelligent than other primates, can use simple tools and have even been taught to communicate using sign language*. They are mainly vegetarian, with the exception of the gorilla, who will occasionally eat small animals.

Research

13 *Use the Internet to research the fascinating story of *'Koko the Gorilla'.*

14 The sloth is a South American mammal that has many features in common with some of the primates we have mentioned. Find out about the sloth and decide whether or not it is a primate.

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